

Complete Puppet Mastery Part-1

Practical 1: Installation and Verification of Puppet Agent on Ubuntu 24.04 (AWS EC2)

▼ Summary

Number of instances | [Info](#)

1

Software Image (AMI)

Ubuntu Server 24.04 LTS (HVM),...[read more](#)

ami-006f82a1d5a27da54

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

 [Preview code](#)

🟢 Success
Successfully initiated launch of instance (i-0717454131f98b6d5)

► Launch log

Next Steps

🔍 What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

```
ubuntu@ip-172-31-12-243: ~
* Support:      https://ubuntu.com/pro

System information as of Mon Jul 27 17:27:32 UTC 2026

System load: 0.0      Temperature:      -273.1 C
Usage of /:  26.6% of 6.71GB  Processes:      110
Memory usage: 25%      Users logged in: 0
Swap usage:  0%      IPv4 address for ens5: 172.31.12.243

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-12-243:~$
```

```
ubuntu@ip-172-31-12-243: ~

System load: 0.0      Temperature:      -273.1 C
Usage of /:  26.6% of 6.71GB  Processes:      110
Memory usage: 25%      Users logged in: 0
Swap usage:  0%      IPv4 address for ens5: 172.31.12.243

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applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-12-243:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
```

```
ubuntu@ip-172-31-12-243: ~  
ubuntu@ip-172-31-12-243:~$ cat /etc/os-release  
PRETTY_NAME="Ubuntu 24.04.4 LTS"  
NAME="Ubuntu"  
VERSION_ID="24.04"  
VERSION="24.04.4 LTS (Noble Numbat)"  
VERSION_CODENAME=noble  
ID=ubuntu  
ID_LIKE=debian  
HOME_URL="https://www.ubuntu.com/"  
SUPPORT_URL="https://help.ubuntu.com/"  
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"  
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"  
UBUNTU_CODENAME=noble  
LOGO=ubuntu-logo  
ubuntu@ip-172-31-12-243:~$ wget https://apt.puppet.com/puppet8-release-noble.deb  
--2026-07-27 17:32:22-- https://apt.puppet.com/puppet8-release-noble.deb  
Resolving apt.puppet.com (apt.puppet.com)... 13.225.5.127, 13.225.5.12, 13.225.5.20, ...  
Connecting to apt.puppet.com (apt.puppet.com)|13.225.5.127|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 3784 (3.7K) [application/vnd.debian.binary-package]  
Saving to: 'puppet8-release-noble.deb'  
  
puppet8-release-noble. 100%[=====>] 3.70K --.-KB/s in 0s  
2026-07-27 17:32:23 (498 MB/s) - 'puppet8-release-noble.deb' saved [3784/3784]  
  
ubuntu@ip-172-31-12-243:~$ sudo dpkg -i puppet8-release-noble.deb  
Selecting previously unselected package puppet8-release.  
(Reading database ... 72608 files and directories currently installed.)  
Preparing to unpack puppet8-release-noble.deb ...  
Unpacking puppet8-release (1.0.0-10noble) ...
```

```
ubuntu@ip-172-31-12-243: ~  
ubuntu@ip-172-31-12-243:~$ sudo dpkg -i puppet8-release-noble.deb  
Selecting previously unselected package puppet8-release.  
(Reading database ... 72608 files and directories currently installed.)  
Preparing to unpack puppet8-release-noble.deb ...  
Unpacking puppet8-release (1.0.0-10noble) ...  
Setting up puppet8-release (1.0.0-10noble) ...  
ubuntu@ip-172-31-12-243:~$ sudo apt update  
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease  
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease  
Get:4 http://apt.puppet.com noble InRelease [56.2 kB]  
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease  
Get:6 http://apt.puppet.com noble/puppet8 amd64 Packages [2221 B]  
Get:7 http://apt.puppet.com noble/puppet8 all Packages [721 B]  
Fetched 59.2 kB in 2s (37.9 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
78 packages can be upgraded. Run 'apt list --upgradable' to see them.  
ubuntu@ip-172-31-12-243:~$ sudo apt install puppet-agent -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following NEW packages will be installed:  
 puppet-agent  
0 upgraded, 1 newly installed, 0 to remove and 78 not upgraded.  
Need to get 36.0 MB of archives.  
After this operation, 125 MB of additional disk space will be used.  
Get:1 http://apt.puppet.com noble/puppet8 amd64 puppet-agent amd64 8.10.0-1noble [36.0 MB  
]  
Fetched 36.0 MB in 0s (89.8 MB/s)
```

```
ubuntu@ip-172-31-12-243: ~  
Get:1 http://apt.puppet.com noble/puppet8 amd64 puppet-agent amd64 8.10.0-1noble [36.0 MB]  
] Fetched 36.0 MB in 0s (89.8 MB/s)  
Selecting previously unselected package puppet-agent.  
(Reading database ... 72613 files and directories currently installed.)  
Preparing to unpack .../puppet-agent_8.10.0-1noble_amd64.deb ...  
Unpacking puppet-agent (8.10.0-1noble) ...  
Setting up puppet-agent (8.10.0-1noble) ...  
Created symlink /etc/systemd/system/multi-user.target.wants/pxp-agent.service → /usr/lib/  
systemd/system/pxp-agent.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/puppet.service → /usr/lib/sy  
stemd/system/puppet.service.  
Removed "/etc/systemd/system/multi-user.target.wants/pxp-agent.service".  
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-12-243:~$ puppet --version  
Command 'puppet' not found, but can be installed with:  
sudo apt install puppet-agent  
ubuntu@ip-172-31-12-243:~$ /opt/puppetlabs/bin/puppet --version  
8.10.0  
ubuntu@ip-172-31-12-243:~$
```

```
ubuntu@ip-172-31-12-243: ~  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-12-243:~$ puppet --version  
Command 'puppet' not found, but can be installed with:  
sudo apt install puppet-agent  
ubuntu@ip-172-31-12-243:~$ /opt/puppetlabs/bin/puppet --version  
8.10.0  
ubuntu@ip-172-31-12-243:~$ /opt/puppetlabs/bin/facter --version  
4.10.0  
ubuntu@ip-172-31-12-243:~$ cat /opt/puppetlabs/bin/facter --version  
cat (GNU coreutils) 9.4  
Copyright (C) 2023 Free Software Foundation, Inc.  
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.  
This is free software: you are free to change and redistribute it.  
There is NO WARRANTY, to the extent permitted by law.  
  
Written by Torbjörn Granlund and Richard M. Stallman.  
ubuntu@ip-172-31-12-243:~$ ls /opt/puppetlabs  
bin facter puppet pxp-agent  
ubuntu@ip-172-31-12-243:~$ /opt/puppetlabs/bin/facter hostname  
ip-172-31-12-243  
ubuntu@ip-172-31-12-243:~$ /opt/puppetlabs/bin/puppet help  
  
Usage: puppet <subcommand> [options] <action> [options]  
  
Available subcommands:  
  
Common:  
agent      The puppet agent daemon  
apply      Apply Puppet manifests locally  
config     Interact with Puppet's settings.  
help       Display Puppet help.
```

```
ubuntu@ip-172-31-12-243: ~  
Available subcommands:  
  
Common:  
agent          The puppet agent daemon  
apply          Apply Puppet manifests locally  
config         Interact with Puppet's settings.  
help           Display Puppet help.  
lookup         Interactive Hiera lookup  
module         Creates, installs and searches for modules on the Puppet Forge.  
resource       The resource abstraction layer shell  
  
Specialized:  
catalog        Compile, save, view, and convert catalogs.  
describe       Display help about resource types  
device         Manage remote network devices  
doc            Generate Puppet references  
epp            Interact directly with the EPP template parser/renderer.  
facts          Retrieve and store facts.  
filebucket     Store and retrieve files in a filebucket  
generate       Generates Puppet code from Ruby definitions.  
node           View and manage node definitions.  
parser         Interact directly with the parser.  
plugin         Interact with the Puppet plugin system.  
script         Run a puppet manifests as a script without compiling a catalog  
ssl            Manage SSL keys and certificates for puppet SSL clients  
  
See 'puppet help <subcommand> <action>' for help on a specific subcommand action.  
See 'puppet help <subcommand>' for help on a specific subcommand.  
Puppet v8.10.0  
ubuntu@ip-172-31-12-243:~$
```

Practical 2: Puppet Development in Isolation Using Local Manifests

```
ubuntu@ip-172-31-12-243: ~/puppet-labs  
node           View and manage node definitions.  
parser         Interact directly with the parser.  
plugin         Interact with the Puppet plugin system.  
script         Run a puppet manifests as a script without compiling a catalog  
ssl            Manage SSL keys and certificates for puppet SSL clients  
  
See 'puppet help <subcommand> <action>' for help on a specific subcommand action.  
See 'puppet help <subcommand>' for help on a specific subcommand.  
Puppet v8.10.0  
ubuntu@ip-172-31-12-243:~$ mkdir ~/puppet-labs  
ubuntu@ip-172-31-12-243:~$ cd ~/puppet-labs  
ubuntu@ip-172-31-12-243:~/puppet-labs$ pwd  
/home/ubuntu/puppet-labs  
ubuntu@ip-172-31-12-243:~/puppet-labs$ nano hello.pp  
ubuntu@ip-172-31-12-243:~/puppet-labs$ cat hello.pp  
file { '/tmp/hello.txt':  
  ensure => file,  
  content => "Hello from Puppet!\n",  
}  
ubuntu@ip-172-31-12-243:~/puppet-labs$ sudo /opt/puppetlabs/bin/puppet apply hello.pp  
Notice: Compiled catalog for ip-172-31-12-243.ap-south-1.compute.internal in environment produ  
ction in 0.01 seconds  
Notice: /Stage[main]/Main/File[/tmp/hello.txt]/ensure: defined content as '{sha256}ffdaae7a2b4  
1b49949d4ed4b04cf76fdaf54e76a8722109e47316f497f7ddce2'  
Notice: Applied catalog in 0.02 seconds  
ubuntu@ip-172-31-12-243:~/puppet-labs$ cat /tmp/hello.txt  
Hello from Puppet!  
ubuntu@ip-172-31-12-243:~/puppet-labs$ sudo /opt/puppetlabs/bin/puppet apply hello.pp  
Notice: Compiled catalog for ip-172-31-12-243.ap-south-1.compute.internal in environment produ  
ction in 0.01 seconds  
Notice: Applied catalog in 0.01 seconds  
ubuntu@ip-172-31-12-243:~/puppet-labs$ cat hello.pp  
file { '/tmp/hello.txt':  
  ensure => file,  
  content => "Hello from Puppet!\n",  
}  
ubuntu@ip-172-31-12-243:~/puppet-labs$ |
```

Crux of today: Infrastructure as Code creates the infrastructure, while Configuration Management configures and continuously maintains that infrastructure in its desired state.

```
ubuntu@ip-172-31-2-124: ~  
sudo: puppet: command not found  
ubuntu@ip-172-31-2-124:~$ puppet apply attributes.pp  
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.01 seconds  
Notice: /Stage[main]/Main/File[/tmp/demo.txt]/ensure: defined content as '{sha256}510e64dae899c2733d854fc29b79812b776c6df86946afffeec64c2d54ebdc6c'  
Notice: Applied catalog in 0.02 seconds  
ubuntu@ip-172-31-2-124:~$ cat /tmp/demo.txt  
This file is managed by me.  
ubuntu@ip-172-31-2-124:~$ ls -l /tmp/demo.txt  
-rw-r--r-- 1 ubuntu ubuntu 28 Jul 30 12:12 /tmp/demo.txt  
ubuntu@ip-172-31-2-124:~$ puppet apply attributes.pp  
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.01 seconds  
Notice: Applied catalog in 0.01 seconds  
ubuntu@ip-172-31-2-124:~$ cat attributes.pp  
file { '/tmp/demo.txt':  
  ensure => file,  
  owner  => 'ubuntu',  
  group  => 'ubuntu',  
  mode   => '0644',  
  content => "This file is managed by me.\n",  
}  
ubuntu@ip-172-31-2-124:~$
```

```
ubuntu@ip-172-31-2-124: ~  
ubuntu@ip-172-31-2-124:~$ nano variables.pp  
ubuntu@ip-172-31-2-124:~$ cat variables.pp  
$file='/tmp/info.txt'  
$owner='ubuntu'  
$group='ubuntu'  
$mode='0644'  
$message="Hello from Variables!\n"  
  
file{$file:  
  ensure=>file,  
  owner=>$owner,  
  group=>$group,  
  mode=>$mode,  
  content=>$message  
}  
ubuntu@ip-172-31-2-124:~$ puppet apply variables.pp  
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.01 seconds  
Notice: /Stage[main]/Main/File[/tmp/info.txt]/ensure: defined content as '{sha256}5709bba439b6ceb26cafd79b83c36798b2b8ac73f2e959fb6913a64474e14fa5'  
Notice: Applied catalog in 0.02 seconds  
ubuntu@ip-172-31-2-124:~$ cat /tmp/info.txt  
Hello from Variables!  
ubuntu@ip-172-31-2-124:~$ ls /tmp/info.txt
```


Practical 3: Advanced Puppet: Understanding and Configuring puppet.conf

```
ubuntu@ip-172-31-2-124: ~  
ubuntu@ip-172-31-2-124:~$ ls -l /etc/puppetlabs/puppet/  
total 16  
drwxr-x--- 2 root root 4096 Jul 30 13:26 devices  
-rw-r--r-- 1 root root 161 Oct 18 2024 hiera.yaml  
-rw-r--r-- 1 root root 434 Jul 31 07:21 puppet.conf  
drwxrwx--x 7 root root 4096 Jul 31 07:21 ssl  
ubuntu@ip-172-31-2-124:~$ cat /etc/puppetlabs/puppet/puppet.conf  
# This file can be used to override the default puppet settings.  
# See the following links for more details on what settings are available:  
# - https://puppet.com/docs/puppet/latest/config_important_settings.html  
# - https://puppet.com/docs/puppet/latest/config_about_settings.html  
# - https://puppet.com/docs/puppet/latest/config_file_main.html  
# - https://puppet.com/docs/puppet/latest/configuration.html  
[agent]  
runinterval = 1800  
ubuntu@ip-172-31-2-124:~$ sudo puppet config print runinterval  
sudo: puppet: command not found  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print runinterval  
1800  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print certname  
ip-172-31-2-124.ap-south-1.compute.internal  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print environment
```

```
ubuntu@ip-172-31-2-124: ~  
1800  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print certname  
ip-172-31-2-124.ap-south-1.compute.internal  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print environment  
production  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print server  
puppet  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print confdir  
/etc/puppetlabs/puppet  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print codedir  
/etc/puppetlabs/code  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print vardir  
/opt/puppetlabs/puppet/cache  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print all  
agent_catalog_run_lockfile = /opt/puppetlabs/puppet/cache/state/agent_catalog_run.lock  
agent_disabled_lockfile = /opt/puppetlabs/puppet/cache/state/agent_disabled.lock  
allow_duplicate_certs = false  
allow_pson_serialization = false  
always_retry_plugins = true  
autoflush = true  
autosign = /etc/puppetlabs/puppet/autosign.conf
```

```
ubuntu@ip-172-31-2-124: ~  
vardir = /opt/puppetlabs/puppet/cache  
vendormoduledir = /opt/puppetlabs/puppet/vendor_modules  
versioned_environment_dirs = false  
waitforcert = 120  
waitforlock = 0  
write_catalog_summary = true  
yamldir = /opt/puppetlabs/puppet/cache/yaml  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config set runinterval  
600 --section agent  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print runinterv  
al --section agent  
600  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/puppet  
cat: /etc/puppetlabs/puppet: Is a directory  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/puppet/puppet.conf  
# This file can be used to override the default puppet settings.  
# See the following links for more details on what settings are available:  
# - https://puppet.com/docs/puppet/latest/config_important_settings.html  
# - https://puppet.com/docs/puppet/latest/config_about_settings.html  
# - https://puppet.com/docs/puppet/latest/config_file_main.html  
# - https://puppet.com/docs/puppet/latest/configuration.html  
[agent]  
runinterval = 600  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config set runinterval
```

```
ubuntu@ip-172-31-2-124: ~  
600  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/puppet  
cat: /etc/puppetlabs/puppet: Is a directory  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/puppet/puppet.conf  
# This file can be used to override the default puppet settings.  
# See the following links for more details on what settings are available:  
# - https://puppet.com/docs/puppet/latest/config_important_settings.html  
# - https://puppet.com/docs/puppet/latest/config_about_settings.html  
# - https://puppet.com/docs/puppet/latest/config_file_main.html  
# - https://puppet.com/docs/puppet/latest/configuration.html  
[agent]  
runinterval = 600  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config set runinterval  
1800 --section agent  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/puppet/puppet.conf  
# This file can be used to override the default puppet settings.  
# See the following links for more details on what settings are available:  
# - https://puppet.com/docs/puppet/latest/config_important_settings.html  
# - https://puppet.com/docs/puppet/latest/config_about_settings.html  
# - https://puppet.com/docs/puppet/latest/config_file_main.html  
# - https://puppet.com/docs/puppet/latest/configuration.html  
[agent]  
runinterval = 1800  
ubuntu@ip-172-31-2-124:~$ |
```


Practical 4: Package Management in Puppet Using the package Resource

```
ubuntu@ip-172-31-2-124: ~
attributes.pp puppet8-release-noble.deb variables.pp
ubuntu@ip-172-31-2-124:~$ dpkg -s tree
dpkg-query: package 'tree' is not installed and no information is available
Use dpkg --info (= dpkg-deb --info) to examine archive files.
ubuntu@ip-172-31-2-124:~$ nano package.pp
ubuntu@ip-172-31-2-124:~$ sudo puppet apply package.pp
sudo: puppet: command not found
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply package.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.32 seconds
Notice: /Stage[main]/Main/Package[tree]/ensure: created
Notice: Applied catalog in 7.23 seconds
ubuntu@ip-172-31-2-124:~$ tree --version
tree v2.1.1 © 1996 - 2023 by Steve Baker, Thomas Moore, Francesc Rocher, Florian Sesser, Kyosuke Tokoro
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply package.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.31 seconds
Notice: Applied catalog in 0.32 seconds
ubuntu@ip-172-31-2-124:~$ cat package.pp
package{'tree':
  ensure=>installed
}
ubuntu@ip-172-31-2-124:~$ nano package.pp
```

```
ubuntu@ip-172-31-2-124: ~
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply package.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.26 seconds
Notice: /Stage[main]/Main/Package[tree]/ensure: created
Notice: Applied catalog in 4.22 seconds
ubuntu@ip-172-31-2-124:~$ curl --version
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 (+libidn2/2.3.7) libssh/0.10.6/openssl/zlib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.11
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtsp rtsp scp sftp smb smbs smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
ubuntu@ip-172-31-2-124:~$ tree --version
tree v2.1.1 © 1996 - 2023 by Steve Baker, Thomas Moore, Francesc Rocher, Florian Sesser, Kyosuke Tokoro
ubuntu@ip-172-31-2-124:~$ cat package.pp
package{'tree':
  ensure=>installed
}
package{'curl':
  ensure=>installed
}
```

```
ubuntu@ip-172-31-2-124: ~  
Processor Puppet::Reports::Store  
ubuntu@ip-172-31-2-124:~$ cat package.pp  
package{['git','curl','tree']:  
  ensure=>absent  
}  
ubuntu@ip-172-31-2-124:~$ puppet parser validate package.pp  
ubuntu@ip-172-31-2-124:~$ sudo puppet apply package.pp --trace  
sudo: puppet: command not found  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply package.pp --tra  
ce  
sudo: /opt/puppetlabs/bin/puppet: command not found  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply package.pp --tra  
ce  
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in envi  
ronment production in 0.24 seconds  
Notice: Applied catalog in 0.32 seconds  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet resource package git  
package { 'git':  
  ensure    => 'absent',  
  provider => 'apt',  
}  
ubuntu@ip-172-31-2-124:~$ puppet --version  
8.10.0  
ubuntu@ip-172-31-2-124:~$
```

Practical 5: Designing and Deploying a Custom Puppet Module in a Simulated Environment

```
MINGW64:/d/downloads
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print modulepath
/etc/puppetlabs/code/environments/production/modules:/etc/puppetlabs/code/modules:/opt/puppetlabs/puppet/modules
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet module generate mymodule
Error: 'module' has no 'generate' action. See 'puppet help module'.
ubuntu@ip-172-31-2-124:~$ cd /etc/puppetlabs/code/environments/production/modules
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ sudo mkdir -p mymodule/{manifests,files,templates,lib,facts.d}
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ ls mymodule
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ sudo nano mymodule/manifests/init.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ tree mymodule
-bash: /usr/bin/tree: No such file or directory
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ ls mymodule
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ cd mymodule
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule$ ls
facts.d files lib manifests templates
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule$ cd manifests
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ ls
```

```
MINGW64:/d/downloads
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule$ cd manifests
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ ls
init.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cat init.pp
class mymodule {
  file { ['/tmp/module-demo.txt']:
    ensure=>file,
    owner=>'ubuntu',
    group=>'ubuntu',
    mode=>'0644',
    content=>"Hello from my first puppet module!\n"
  }
}
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cd ..
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule$ cd ..
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ cd ~
ubuntu@ip-172-31-2-124:~$ nano site.pp
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet parser validate site.pp
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.02 seconds
Notice: /Stage[main]/Mymodule/File[/tmp/module-demo.txt]/ensure: defined content
```

```
MINGW64:/d/downloads
~
ubuntu@ip-172-31-2-124:~$ nano site.pp
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet parser validate site.p
p
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in envi
ronment production in 0.02 seconds
Notice: /Stage[main]/Mymodule/File[/tmp/module-demo.txt]/ensure: defined content
as '{sha256}783e3f58119087a5400e3a6ba8a996f7ebbebacc39148f9fe37dbb8a9db5a6d'
Notice: Applied catalog in 0.01 seconds
ubuntu@ip-172-31-2-124:~$ ls -l /tmp/module-demo.txt
-rw-r--r-- 1 ubuntu ubuntu 35 Jul 31 09:43 /tmp/module-demo.txt
ubuntu@ip-172-31-2-124:~$ cat /tmp/module-demo.txt
Hello from my first puppet module!
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in envi
ronment production in 0.02 seconds
Notice: Applied catalog in 0.01 seconds
ubuntu@ip-172-31-2-124:~$
Broadcast message from root@ip-172-31-2-124 (Fri 2026-07-31 09:50:43 UTC):

The system will power off now!

Connection to ec2-13-233-154-135.ap-south-1.compute.amazonaws.com closed by remo
te host.
Connection to ec2-13-233-154-135.ap-south-1.compute.amazonaws.com closed.

sreev@admin MINGW64 /d/downloads
$ |
```

Practical 6: Developing Advanced Puppet Modules Using Multiple Classes and Resource Organization

```

ubuntu@ip-172-31-2-124: ~
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cat init.pp
class mymodule {
  include mymodule::install
  include mymodule::config
  include mymodule::service
}
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cat install.pp
class mymodule::install{
  package{'tree':
    ensure=>installed,
  }
}
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cat config.pp
class mymodule::config{
  file{'/tmp/module-demo.txt':
    ensure=>file,
    owner=>'ubuntu',
    group=>'ubuntu',
    mode=>'0644',
    content=>"Configured by advanced puppet module\n"
  }
}
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cat service.pp
class mymodule::service{

```

```

  service{'ssh':
    ensure=>running,
    enable=>true,
  }
}
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cd ~
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.36 seconds
Notice: Applied catalog in 0.35 seconds
ubuntu@ip-172-31-2-124:~$ tree --version
tree v2.1.1 © 1996 - 2023 by Steve Baker, Thomas Moore, Francesc Rocher, Florian Sesser, Kyosuke Tokoro
ubuntu@ip-172-31-2-124:~$ cat /tmp/module-demo.txt
Configured by advanced puppet module
ubuntu@ip-172-31-2-124:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-31 11:19:49 UTC; 46min ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 771 (sshd)
     Tasks: 1 (limit: 627)
    Memory: 5.1M (peak: 7.1M)
       CPU: 78ms
    CGroup: /system.slice/ssh.service
            └─771 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

```


Practical 7 - Developing Parameterized and Reusable Puppet Modules

```
ubuntu@ip-172-31-2-124: ~  
Last login: Fri Jul 31 11:19:56 2026 from 128.185.168.206  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet config print modulepath  
/etc/puppetlabs/code/environments/production/modules:/etc/puppetlabs/code/modules:/opt/puppetlabs/puppet/modules  
ubuntu@ip-172-31-2-124:~$ cd /etc/puppetlabs/code/environments/production/modules  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ cd manifests  
-bash: cd: manifests: No such file or directory  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ ls  
mymodule  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ cd mymodule  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule$ cd manifests  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ ls  
config.pp  init.pp  install.pp  service.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ ls  
config.pp  init.pp  install.pp  service.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ sudo nano init.pp
```

```
ubuntu@ip-172-31-2-124: ~  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ sudo nano config.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cd ~  
ubuntu@ip-172-31-2-124:~$ sudo nano site.pp  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet parser validate site.pp  
Error: Could not parse for environment production: Syntax error at '=' (file: /home/ubuntu/site.pp, line: 4, column: 17)  
ubuntu@ip-172-31-2-124:~$ sudo nano site.pp  
ubuntu@ip-172-31-2-124:~$ sudo nano site.pp  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet parser validate site.pp  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/init.pp  
Error: Could not parse application options: invalid option: /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/init.pp  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet parser validate /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/init.pp  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet parser validate /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/config.pp  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp  
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.51 seconds
```

```

ubuntu@ip-172-31-2-124: ~
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.51 seconds
Error: Parameter path failed on File[/tmp/module-demo.txt]: File paths must be fully qualified, not 'tmp/module-demo.txt' (file: /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/config.pp, line: 2)
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet parser validate /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/config.pp
ubuntu@ip-172-31-2-124:~$ sudo nano /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/config.pp
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.38 seconds
Notice: /Stage[main]/Mymodule::Config/File[/tmp/module-demo.txt]/ensure: defined content as '{sha256}252a3b2fbf34b6c8a6294971cd08ddf11cd8f253dc31dd8da6acfaf59ee8e22'
Notice: Applied catalog in 0.38 seconds
ubuntu@ip-172-31-2-124:~$ cat /tmp/module-demo.txt
Message : Welcome to Puppet Modules
Port : 8080
ubuntu@ip-172-31-2-124:~$ cat site.pp
class{'mymodule':
  message=>"Welcome to Puppet Modules",
  port=>8080,
  manage_service=>true
}

```

```

ubuntu@ip-172-31-2-124: ~
ubuntu@ip-172-31-2-124:~$ sudo nano site.pp
ubuntu@ip-172-31-2-124:~$
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.43 seconds
Notice: /Stage[main]/Mymodule::Config/File[/tmp/module-demo.txt]/content: content changed '{sha256}252a3b2fbf34b6c8a6294971cd08ddf11cd8f253dc31dd8da6acfaf59ee8e22' to '{sha256}cc1796f600443ddd5d3ed0a87a485eccb5504788ecf06e71514f843fc1bf196d'
Notice: Applied catalog in 0.38 seconds
ubuntu@ip-172-31-2-124:~$ cat /tmp/module-demo.txt
Message : Database Server
Port : 3306
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/config.pp
class mymodule::config(String $message,Integer $port){
  file{'/tmp/module-demo.txt':
    ensure=>file,
    owner=>'ubuntu',
    group=>'ubuntu',
    mode=>'0644',
    content=>"Message : ${message}\nPort : ${port}\n"
  }
}

```

```

ubuntu@ip-172-31-2-124: ~
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/code/environments/production/
modules/mymodule/manifests/init.pp
class mymodule(String $message="Configured by puppet",
  Integer $port=80,
  Boolean $manage_service=true){
  include mymodule::install
  class{'mymodule::config':
    message=>$message,
    port=>$port
  }
  if $manage_service{
    include mymodule::service
  }
}

ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/code/environments/production/
modules/mymodule/manifests/install.pp
class mymodule::install{
  package{'tree':
    ensure=>installed,
  }
}

ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/code/environments/production/
modules/mymodule/manifests/service.pp

```

Practical 8 - Developing Production-Style Puppet Modules Using EPP Templates and Parameterized Classes

```

sreev@admin MINGW64 ~ (main)
$ cd d:

sreev@admin MINGW64 /d
$ cd downloads

sreev@admin MINGW64 /d/downloads
$ ls

sreev@admin MINGW64 /d/downloads
$ ssh -i "puppet-lab-key.pem" ubuntu@ec2-15-206-88-79.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-15-206-88-79.ap-south-1.compute.amazonaws.com (15.206.88.79)' can't be established.
ED25519 key fingerprint is SHA256:3N3FQ6Q3CjJ4uMiOdHJ0SKQiCu7NgffBpeuSeSa6AnE.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:20: ec2-13-201-18-208.ap-south-1.compute.amazonaws.com
  ~/.ssh/known_hosts:23: ec2-13-204-76-202.ap-south-1.compute.amazonaws.com
  ~/.ssh/known_hosts:24: ec2-15-206-210-61.ap-south-1.compute.amazonaws.com
  ~/.ssh/known_hosts:25: ec2-13-233-154-135.ap-south-1.compute.amazonaws.com
  ~/.ssh/known_hosts:26: ec2-35-154-5-94.ap-south-1.compute.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

```

```

ubuntu@ip-172-31-2-124: ~
ubuntu@ip-172-31-2-124:~$ cd /etc/puppetlabs/code/environments/production/modules/mymodule/templates
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/templates$ sudo nano config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/templates$ cd ..
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule$ cd manifests
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ sudo nano config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ sudo nano init.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cd ..
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule$ cd manifests
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ sudo /opt/puppetlabs/bin/puppet parser validate config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ sudo /opt/puppetlabs/bin/puppet parser validate init.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/manifests$ cd ~
ubuntu@ip-172-31-2-124:~$ sudo nano site.pp
ubuntu@ip-172-31-2-124:~$ puppet parser validate site.pp

```

```

ubuntu@ip-172-31-2-124: ~
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp
Error: Evaluation Error: Error while evaluating a Function Call, Could not find template 'mymodule/config.epp' (file: /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/config.pp, line: 8, column: 14) on node ip-172-31-2-124.ap-south-1.compute.internal
ubuntu@ip-172-31-2-124:~$ cd /etc/puppetlabs/code/environments/production/modules/mymodule/templates
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/templates$ ls
config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/templates$ sudo nano config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/templates$ ls
config.epp  config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/templates$ sudo rm config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/templates$ ls
config.epp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/mymodule/templates$ cd ~
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in envi

```

```
ubuntu@ip-172-31-2-124: ~  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp  
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in envi  
ronment production in 0.38 seconds  
Notice: /Stage[main]/Mymodule::Config/File[/tmp/module-demo.txt]/content: conten  
t changed '{sha256}cc1796f600443ddd5d3ed0a87a485eccb5504788ecf06e71514f843fc1bf1  
96d' to '{sha256}f46a0d720e85bd2d2e289f686f2e0a5deb72ed133ac395a5cfec221e9b92329  
e'  
Notice: Applied catalog in 0.37 seconds  
ubuntu@ip-172-31-2-124:~$ cat /tmp/module-demo.txt  
-----  
Puppet configuration report  
  
Server message: Production Web server  
Listening port: 8080  
Environment: Production  
Hostname: web01Vishnu  
Generated by: Puppet EPP template  
  
-----  
ubuntu@ip-172-31-2-124:~$ sudo nano site.pp  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp  
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in envi  
ronment production in 0.39 seconds  
Notice: /Stage[main]/Mymodule::Config/File[/tmp/module-demo.txt]/content: conten
```

```
ubuntu@ip-172-31-2-124: ~  
Generated by: Puppet EPP template  
  
-----  
ubuntu@ip-172-31-2-124:~$ sudo nano site.pp  
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp  
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in envi  
ronment production in 0.39 seconds  
Notice: /Stage[main]/Mymodule::Config/File[/tmp/module-demo.txt]/content: conten  
t changed '{sha256}f46a0d720e85bd2d2e289f686f2e0a5deb72ed133ac395a5cfec221e9b923  
29e' to '{sha256}2becde3fb139b6dcd627749c20df09036189905c920e9d7e4ee42277325f332  
6'  
Notice: Applied catalog in 0.37 seconds  
ubuntu@ip-172-31-2-124:~$ cat /tmp/module-demo.txt  
-----  
Puppet configuration report  
  
Server message: Database server  
Listening port: 3306  
Environment: Testing reusability  
Hostname: db01Vishnu  
Generated by: Puppet EPP template  
  
-----  
ubuntu@ip-172-31-2-124:~$ |
```



```
ubuntu@ip-172-31-2-124: ~  
ubuntu@ip-172-31-2-124:~$ cat /etc/puppetlabs/code/environments/production/modules/mymodule/templates/config.epp  
-----  
Puppet configuration report  
  
Server message: <%= $message %>  
Listening port: <%= $port %>  
Environment: <%= $environment %>  
Hostname: <%= $hostname %>  
Generated by: Puppet EPP template  
  
-----  
ubuntu@ip-172-31-2-124:~$ cat /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/init.pp  
class mymodule(String $message="Configured by puppet",  
  Integer $port=80,  
  Boolean $manage_service=true,  
  String $hostname="localhost",  
  String $environment="Development"){  
  include mymodule::install  
  class{'mymodule::config':  
    message=>$message,  
    port=>$port,  
    environment=>$environment,
```

```
  }  
  if $manage_service{  
    include mymodule::service  
  }  
}  
  
ubuntu@ip-172-31-2-124:~$ cat /etc/puppetlabs/code/environments/production/modules/mymodule/manifests/config.pp  
class mymodule::config(String $message,Integer $port,String $environment,  
String $hostname){  
  file{'/tmp/module-demo.txt':  
    ensure=>file,  
    owner=>'ubuntu',  
    group=>'ubuntu',  
    mode=>'0644',  
    content=>epp(  
      'mymodule/config.epp',  
      {  
        'message'=>$message,  
        'port'=>$port,  
        'environment'=>$environment,  
        'hostname'=>$hostname  
      }  
    )  
  }  
}
```